

# HW/Lab 3: Multipath Channel and OFDM

## ECSE 4560/6560: Modern Communication Systems

### RPI Fall 2025

## Goals and Instructions

The goals for this assignment are:

- Understand how multipath delay spread influences OFDM symbol design and cyclic prefix overhead.
- Derive conditions for orthogonality of OFDM subcarriers and the required subcarrier spacing.
- Explore the resource grid of 5G New Radio (NR) and compute maximum throughput for a given numerology.
- Implement and test OFDM channel estimation in MATLAB, including effects of CP, guard bands, DC subcarrier, SNR, and timing offsets.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code.
- **Tools:** We will primarily use MATLAB (and/or Python). Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Submission:** upload a single PDF with your solutions and MATLAB code to Gradescope before the deadline. You may use MATLAB's **Publish** feature to generate a compiled PDF and attach it as your single submission. Include your name on the PDF.

## Theory Problems

### 1. OFDM Theory

An OFDM system operates at a carrier frequency of 2.0 GHz with a total bandwidth of 5 MHz. The channel has a maximum delay spread of  $\tau_{\max} = 4 \mu\text{s}$ . Design the OFDM numerology such that the cyclic prefix overhead is 10% (i.e., 90% efficiency). Assume that the entire 5 MHz bandwidth is used for subcarriers, without allocating guard bands.

- a) Determine the useful OFDM symbol duration  $T_{\text{ofdm}}$  and cyclic prefix duration  $T_{\text{cp}}$ .

- b) Find the subcarrier spacing  $\Delta f$  and calculate the number of subcarriers  $N$  required.
- c) Suggest a practical FFT size (power of two) that would be chosen in an implementation.
- d) (ECSE 6560 only) Now repeat the design for the case when the OFDM system supports mobility up to 360 km/h. Compute the maximum Doppler shift at  $f_c = 2.0$  GHz and estimate the corresponding channel coherence time.
- e) (ECSE 6560 only) Based on the coherence time, propose a reference signal (pilot) design that maintains 90% efficiency, and illustrate the pilot arrangement with a labeled time-frequency grid figure (showing OFDM symbols on the time axis and subcarriers on the frequency axis).

## 2. OFDM Orthogonality

In OFDM, data symbols are mapped onto closely spaced subcarriers that remain mutually orthogonal over the useful symbol interval. Consider the time-limited complex exponentials

$$b_k(t) = \begin{cases} e^{j2\pi f_k t}, & 0 \leq t \leq T_s, \\ 0, & \text{otherwise,} \end{cases} \quad k \in \{0, 1, \dots, N-1\},$$

and the OFDM signal

$$S(t) = \frac{1}{N} \sum_{k=0}^{N-1} d_k b_k(t),$$

where  $d_k$  are data symbols (e.g., BPSK/QPSK/QAM). Define the inner product

$$\langle b_i, b_j \rangle = \frac{1}{T_s} \int_0^{T_s} b_i(t) b_j^*(t) dt.$$

Answer the following:

- a) For which condition on  $i, j$  is  $\langle b_i, b_j \rangle = 1$ ?
- b) Let  $\Delta f = f_i - f_j$ . For which condition on  $f_i, f_j$  is  $\langle b_i, b_j \rangle = 0$ ?
- c) (ECSE 6560 only) What is the *minimum* subcarrier spacing  $\Delta f_{\min}$  that ensures orthogonality for a given  $T_s$ ?

*Hint:* You may find the integral  $\int_0^{T_s} e^{j2\pi\Delta f t} dt$  useful.

## 3. OFDM Example with 5G New Radio

A 5G NR base-station is operating with the following specifications:

- **Time-domain structure:** A radio frame has a duration of 10 ms and is divided into 10 subframes, each of 1 ms. Each subframe contains  $2^\mu$  slots, where  $\mu$  is the numerology index. For subcarrier spacing  $\Delta f = 30$  kHz, we have  $\mu = 1$ , so each subframe contains 2 slots. Each slot consists of 14 OFDM symbols.
  - **Frequency-domain structure:** The system uses 1272 subcarriers, each spaced 30 kHz apart. Assume all subcarriers are active for data transmission.
  - **Modulation:** Each subcarrier is modulated using QPSK.
  - **Simplifying assumption:** Ignore control channels, reference signals, or guard bands (i.e., assume 100% of the grid is used for data).
- a) Draw a 2D time-frequency resource grid (time on the  $x$ -axis, subcarriers on the  $y$ -axis). Clearly label the hierarchy: frames, subframes, slots, OFDM symbols, and subcarriers.

- b) Calculate the maximum theoretical throughput (in bits/s) under the above assumptions. Proceed as follows:
- Determine the number of bits carried by one subcarrier with QPSK modulation.
  - Calculate the number of bits per OFDM symbol (using all 1272 subcarriers).
  - Find the total number of OFDM symbols transmitted per second.
  - Compute the final throughput in bits/s.

## Matlab Simulations: Pilot-Only OFDM Channel Estimation

**Learning goals.** In this exercise you will (i) simulate a tapped-delay multipath channel acting on a single OFDM symbol that carries only pilots, (ii) add AWGN, (iii) implement the OFDM receiver front end (CP removal + FFT), (iv) perform per-subcarrier least-squares (LS) channel estimation  $\hat{H}[k] = Y[k]/X[k]$ , and (v) compare the *true* channel vs. the *estimated* channel in both frequency and time domains. You will also study the effects of guard bands, DC-subcarrier nulling, SNR, CP length, and small timing offsets within the CP.

### Starter code and scope

You are given a starter MATLAB script (On Piazza) that constructs the pilot grid in the *frequency domain*, performs IFFT and CP insertion, and computes the *true* channel responses  $H_{\text{true}}[k]$  and  $h[n]$ . Complete the *receiver side* and experiments as described below (blocks marked TODO in the starter). **Keep the use of `fftshift`/`ifftshift`** as in the starter code.

### Default parameters (unless a subpart says otherwise):

- FFT size  $N = 64$ , CP length  $\text{CP} = 24$ ,  $\text{SNR} \in \{30 \text{ dB}, 0 \text{ dB}\}$ .
- Subcarrier mapping: randomized  $\pm 1$  pilots on used tones. **Defaults:** `useGuards=false`, `guard=10`, `useDC=true`.
- Channel: tapped delay line with delays `[0 2 5 9 13]` (Delay spread  $L_h = 14$  samples) and complex gains normalized to unit power (as in the provided code).
- Frequency-domain truth:  $H_{\text{true}}[k]$  computed as the  $N$ -point DFT of zero-padded  $h[n]$ , plotted in *shifted* order.
- Time-domain estimate:  $\hat{h}[n]$  obtained from the IDFT of  $\hat{H}[k]$  (circular impulse response).

### What you must implement

- Apply the channel and AWGN.** Convolve the CP-prefixed time signal with  $h[n]$  (linear convolution), then add AWGN (additive white Gaussian noise). Feel free to use matlab's `awgn()` function, but make sure to pass the '`measured`' parameter inside `awgn` (see documentation of `awgn()` function).
- CP removal and FFT.** Extract the  $N$  samples immediately after CP, then take an  $N$ -point FFT to obtain  $Y[k]$  (keep unshifted for math, use `fftshift` for plotting consistency with the given axes).
- Per-subcarrier LS estimate.** Compute channel estimate  $\hat{H}[k] = Y[k]/X[k]$ ; set  $\hat{H}[k] = 0$  on nulled tones (DC/guards) for plotting.

## Deliverables (plots & metrics)

Unless a subpart specifies a variation, include the following **for each experiment**:

1. **Frequency domain (shifted)**: Magnitude and phase of  $H_{\text{true}}[k]$  and  $\hat{H}[k]$  vs. subcarrier index.
2. **Time domain**: Magnitude and phase of  $h[n]$  and  $\hat{h}[n]$  vs. time index.
3. **Scalar metric** (report in title of the plot) NMSE over used tones,

$$\text{NMSE} = \frac{\sum_{k \in \mathcal{U}} |\hat{H}[k] - H_{\text{true}}[k]|^2}{\sum_{k \in \mathcal{U}} |H_{\text{true}}[k]|^2},$$

where  $\mathcal{U}$  is the set of used (nonzero-pilot) subcarriers.

## Experiments and analysis (answer all subparts with channel plots and explanation)

- a) **Baseline (clean channel estimate)**. Use the default parameters (`CP=24`, `useGuards=false`, `useDC=true`, `SNRdB=30`). Produce all deliverable plots and the NMSE. Briefly explain why  $\hat{H}[k]$  closely matches  $H_{\text{true}}[k]$  when  $\text{CP} \geq L_h - 1$  and timing is aligned.
- b) **Effect of DC subcarrier nulling**. Toggle `useDC=false`  $\rightarrow$  `useDC=true` and repeat the baseline. Discuss: (i) why the DC bin is typically nulled in practice, (ii) what you observe on the DC tone in magnitude/phase when enabled vs. nulled, and (iii) the impact (if any) on NMSE (computed only on used tones).
- c) **Guard bands**. Set `useGuards=true` with `guard=10`. Repeat the baseline. Explain the visual difference in spectra (which bins are nulled), how guards affect spectral leakage to adjacent channels (conceptually), and comment on any changes to NMSE on the remaining used tones.
- d) **Noise stress test**. Run the baseline with `SNRdB=0`. Compare  $|H_{\text{true}}[k]|$  vs.  $|\hat{H}[k]|$  and the phase plots. Explain how noise perturbs the Least Square channel estimate. Report the NMSE at 0 dB vs. 30 dB and relate to your plots.
- e) **Too-short CP (ISI)**. Set `CP=2` (keep `SNRdB=30`, guards/DC as in the baseline). Repeat the deliverables. Explain why the per-tone model  $Y[k] = H[k]X[k]$  breaks when  $\text{CP} < L_h - 1$ , how that manifests in your frequency plots (distortion across subcarriers) and in time domain (smeared/wrapped  $\hat{h}[n]$ ), and quantify the NMSE increase.
- f) **Timing misalignment within CP**. Keep `CP=24`, `SNRdB=30`. Emulate a receiver timing offset of +4 samples *inside the CP* by slicing `y_rx(CP+1+4 : CP+N+4)`. Compare with the aligned baseline. Show that  $\hat{H}[k]$  exhibits (approximately) a linear phase ramp across  $k$  (explain why), while magnitudes remain close; comment on NMSE.

## Deliverables

- **One PDF** produced via MATLAB *Publish* (recommended): include your code cells *and* all figures with captions and brief answers to each subpart (a)–(f). Clearly label each experiment.